

Umer Irfan

Lahore · Pakistan

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EDUCATION

Lahore University of Management Sciences

BSc Computer Science · Minor in Computer Engineering

Sept 2023 to July 2027

Pak-Turk Maarif International Schools and Colleges

A-Levels: 3A/A* · O-Levels: 10A*

CS GPA **3.707/4.00** · CGPA **3.491/4.00**

June 2023

RESEARCH INTERESTS

Embedded Systems · Hardware Security · Firmware Security · Computer Architecture · Operating Systems

HONOURS AND AWARDS

- **1st Place.** *Pakistan Semiconductor Summit Hackathon* 2026. RISC-V RTL verification with Sail and Verilator.
- **Best Project.** Intel ISEF Pakistan 2019.
- **Dean's Honour List.** LUMS. Three semesters.
- **Team Pakistan.** *International Young Physicists Tournament* in Beijing 2018.
- **Best in Category.** Teknofest 2020 in Gaziantep Turkey.

PUBLICATIONS

Accepted Publications

- ***Motion-Coupled Sensing: Self-Powered State-Change Sensing*** Co-author · *IEEE MASS 2026*
 - Co-developed **batteryless sensing** in which hinge motion powered a complete sensing and LoRa transmission cycle. Removed periodic polling and battery replacement.
 - Designed **ATmega328P hardware** and LoRa firmware. Cut a firmware loop from **520 to 7 CPU cycles** using fixed-point arithmetic.
 - Built a **stepper-driven stress-test rig** for **10,000 actuations**. The system achieved **99.3% transmission reliability** over 5,945 bin-lid actuations at five sites.
- ***Kissan-Dost: Conversational IoT for Precision Agriculture*** Co-author · *IEEE DCOSS-IoT 2026*
 - Co-developed **Urdu WhatsApp text and voice advice** so smallholder farmers could act on live sensor readings without interpreting dashboards.
 - Built a **multi-hop ESP-NOW monitoring mesh** and ESP32 gateway for soil and climate sensing.
 - Deployed at **two field sites for 45 days each**. A five-participant pilot found near-daily chatbot use and irrigation decisions informed by the guidance.

Submitted Manuscript

- ***PARDA: On-Device Audio Privacy for Smart Glasses*** First author · *IEEE PerCom 2027*
 - Designed **context-aware redaction and voice replacement** to reduce bystander disclosure across conversation turns, beyond explicit personal identifiers.
 - Built a silver-labeling pipeline for **717 CANDOR conversations**. Distilled a retrieval-aware **2B model** with preference optimization. Deployed Q4 inference on **Raspberry Pi 5**.
 - Reduced model-judged leakage from **0.502 to 0.378** on **143 held-out conversations** while utility rose from 0.875 to 0.900. Measured a **0.77 weighted real-time factor** for the audio pipeline alone.

Manuscripts in Preparation

- ***Programmer-in-the-Middle: Firmware Implantation via Flash Programmers*** Joint first author · *Pre-Release*
 - Demonstrated a **firmware supply-chain trust gap**. Compromised flash programmers altered target code after compilation while returning clean host readback checks.
 - Modified USBasp firmware for **AVR reset-vector hijacking** and stub injection. Spoofed host readback checks.
 - **Reverse-engineered ST-Link V2** firmware and tested custom programmer firmware across **three target boards**.
- ***On-Chip Behavioral Monitoring for Runtime Hardware Trojan Detection*** Joint first author · *Pre-Release*
 - Designed **transaction-level monitoring** for runtime detection of hardware Trojans that evade pre-deployment tests.
 - Built a trace-collection pipeline for expected and **Trojan-affected execution** across computational and application workloads. Developed a **1D CNN classifier**.
 - Specified an **FPGA monitor architecture**. Planned generalization studies across processor and cache configurations with latency and resource evaluation.

PUBLICATIONS (CONTINUED)

Preprints

- **LLM-Enhanced Wearables for Health Guidance in LMICs** Co-author · [arXiv:2602.08701](#)
 - Co-developed **plain-language WhatsApp health feedback** to make low-cost wearable data understandable to users with limited health literacy.
 - Designed a **\$12.72 screenless ATmega328P band** and BLE firmware. Combined PPG with temperature and motion sensing.
 - Supported a **20-participant 96-hour study**. Mean self-rated health-data comprehension rose from **3.05 to 3.60/5**.
- **WatchTower: Transformer-Based Open-Set Network Anomaly Detection** Joint first author
 - Developed **open-set intrusion detection** for edge devices that must identify attacks absent from their training data.
 - Combined a transformer with **28 packet-level features** and PROSER. Optimized INT8 inference with TVM on Raspberry Pi 5.
 - Reached **3.82 ms inference latency** and **93.66% classification accuracy** on the deployed model.

SELECTED RESEARCH AND TECHNICAL PROJECTS

PixelPacker | FPGA-Accelerated Neural Image Compression 2025 to 2026

- Built edge-cloud image compression for wildlife camera traps on **bandwidth-constrained links**. Exploited fixed backgrounds to avoid retransmitting unchanged content.
- Developed an **Artix-7 FPGA accelerator** with pipelined DSPs, FIFO buffering, and DDR memory access. Implemented **background-aware latent differencing** on ESP32.
- Distilled and INT8-quantized a depthwise-separable encoder from **28.4 MB to 800 kB**. Reduced its memory footprint by approximately **36-fold** for edge deployment.

RIZZ-8 | Custom 8-Bit RISC Processor 2025

- Independently designed and built an **8-bit Harvard CPU** from 74-series ICs on **eight custom PCBs** with a 16-bit address space.
- Defined a **custom ISA** with signed and unsigned arithmetic and bitwise operations.
- Wrote a **Python assembler** and loaded machine code into EEPROM. Executed custom assembly programs on the completed processor.

KeySiphon | Power-Rail Side-Channel Analysis 2025

- Built an instrumented **RSA side-channel prototype** on ATmega328P to study shared-supply leakage. Isolated the supply pin and captured traces through a shunt resistor.
- Trained a **1D CNN** to distinguish exponent-bit operations with **93.10% validation accuracy**. Automated inference from noisy traces.

CARDY | ESP8266-Based Interactive Challenge Platform 2023

- Developed **ESP8266 firmware** and a **custom PCB** for an interactive hardware competition. Integrated an HTTP server with LittleFS and physical controls.
- Coordinated assembly and testing of **50 physical devices**. Supported **50 participating teams** with real-time telemetry.

RESEARCH EXPERIENCE

Research Assistant | **SYSNET Lab** at LUMS May 2024 to Sept 2026

Dr. Naveed Anwar Bhatti

- Built an arbitrary **voltage-trace replay bench** for repeatable CheckMate power experiments using laptop-controlled function generators and amplifiers.
- Built a **5 to 70°C aluminum Peltier chamber** to characterize temperature-dependent device performance.
- **Designed and hand-assembled** a **four-layer nRF52 wristband** for maternal-health temperature monitoring in rural Sindh. Targeted four months per charge.
- Co-developed a **lossless 16-bit thermal codec** using temporal and camera-noise predictors. Files averaged **15% smaller than FFV1** at comparable processing cost, with no larger outputs in the tested sets and a best case of **one-third the size**.

RESEARCH EXPERIENCE (CONTINUED)

Research Assistant | *CHASE Lab* at LUMS

2025 to 2026

Dr. Muhammad Ali Siddiqi

- Developed **FPGA datapaths** for ML inference and characterized **power side-channel leakage** on embedded platforms.
- Assessed a laboratory **pacemaker (IMD)** using **HackRF** and ESP32. Crafted BLE attempts extended observed wake time from roughly **2 to 10.8 seconds**. Measured 1.56 to 1.93 mJ per connection to characterize battery-depletion risk.
- Contributed **FPGA acceleration**, programmer-firmware analysis and transaction-trace collection to PixelPacker, PITM and runtime Trojan monitoring.

TEACHING EXPERIENCE

EE320 Computer Organization & Assembly Language

Sole TA · 2026. Six quizzes. Seven homework sets.

AI624 AI on Edge Devices

TA · 2026. 82 graduate students.

ENGG100 Engineering Design Fundamentals

TA · 2024. 30 engineering freshmen.

CS370 Operating Systems

TA · 2026. 295 students. Kernel and shell assignments.

CS210 Data Structures

TA · 2025. Programming assignments and automated grading.

Diploma in Digital Health

Course Designer · 2026. IoMT classes for 90+ medical professionals.

SELECTED GRADUATE COURSEWORK

- Internet of Things
- Applied Cryptography
- Topics in Computer and Network Security
- AI on Edge Devices
- Computer Architecture
- Distributed Systems

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Call Agent Application

Aug 2024 to March 2025

- Built a **streaming voice agent** using Node.js and WebSockets. Integrated WebRTC VAD with Whisper and GPT-4.
- Cut response latency to **under four seconds** through audio buffering and pipeline optimization.

DevOps Engineer | *U.n.I Social App*

May to July 2024

- Built GitHub Actions workflows for containerized deployment and automated application delivery.
- Configured EC2 infrastructure and NGINX reverse proxies for traffic and load management.

React Native Engineer | Project X

Nov 2024 to June 2025

- Developed an Android application for a smart security-camera startup.
- Implemented **IoT device onboarding**, Firebase authentication, cloud messaging and backend API integration.

TECHNICAL SKILLS

Programming & HDL

C · C++ · Python · x86, MIPS & RISC-V Assembly · Verilog · SystemVerilog

Microcontrollers

PIC16F876A/877A · ATmega328P · ATmega32U2/8U2 · ATtiny85/16 · STM32 · TM4C123GH6PM · nRF52832 · ESP32/ESP8266

FPGA & Architecture

Basys 3 (Artix-7 XC7A35T-1CPG236C) · Nexys A7 (Artix-7 XC7A100T/XC7A50T) · RISC-V · CPU & ISA design · Sail · Verilator

Embedded Systems

SPI · I²C · UART · BLE · ESP-NOW · LoRa · FreeRTOS

Firmware Security

USBasp · ST-Link V2 (SWD) · Ghidra · Binary rewriting · Readback spoofing

Hardware Instrumentation

Oscilloscope · Function generators · HackRF/SDR · Power-trace capture

PCB & Mechanical Design

KiCad · EasyEDA Pro · Eagle · JLCPCB · PCB etching · SolidWorks · Blender

Edge Inference

PyTorch · TensorFlow Lite Micro · TVM · INT8 quantization · Knowledge distillation · FedProx